Eutric Nitisols:

Eutric Nitisols are deep, well-drained tropical soils with a reddish color caused by iron oxide accumulation and a clayey nitric subsurface horizon with strong, polyhedric blocky structure. They are named **“Eutric”** because they have a relatively high base saturation (>50%), which makes them more fertile and less acidic than Dystric Nitisols. Physically stable with a porous, deep solum, they allow unrestricted deep rooting and are highly productive when properly managed.

Drainage:

- **Wet season:** Well-drained; water infiltrates easily due to stable structure.
- **Dry season:** Soil remains friable and workable; moderate moisture retention supports crop growth.
- **Landscape:** Common on undulating to hilly tropical terrain; rarely found in depressions.

Depth:

- Very deep (>150 cm).
- Rooting depth is generally unrestricted due to stable structure.

Workability:

- **When wet:** Plastic and sticky but manageable; does not smear or compact like Vertisols.
- **When dry:** Firm, friable, and easy to till; seedbed preparation is straightforward.

Nutrient Richness & Cation Exchange Capacity (CEC):

- CEC: Moderate to high (15–35 cmol(+)/kg).
- Base saturation is relatively high (>35%), making these soils more fertile than Dystric Nitisols.
- Nitrogen and organic matter are moderate.
- Phosphorus availability is low due to high P-fixation.
- Iron oxides dominate; micronutrients such as zinc and manganese may be sufficient for crops.

pH:

- Slightly acidic to near neutral (5.5–6.8).

Management:

- Highly productive soils if managed well; deep, porous solum allows deep rooting and provides resistance to erosion.
- Good workability, internal drainage, and fair water-holding capacity support both plantation crops and food crops.
- Commonly planted with cocoa, coffee, rubber, pineapple, and various smallholder food crops.
- High phosphorus fixation requires careful P-fertilizer management: apply slow-release, low-grade rock phosphate (several tons/ha every few years) combined with smaller doses of soluble superphosphate for short-term crop response.
- Incorporate organic matter (manure, compost, crop residues) to improve soil fertility, structure, and biological activity.
- Maintain vegetation cover or cover crops on slopes to reduce erosion.
- Liming may be applied if soil acidity increases beyond optimal levels for the crop.